VLAN Configuration and Inter-Switch Connectivity

Cisco Packet Tracer Lab Report

Course	Computer Networking Laboratory
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Institution	Southeast University (SEU), Dhaka
Tool Used	Cisco Packet Tracer
Topic	VLAN Creation, Port Assignment, Trunking & Connectivity Verification

Objective

This lab demonstrates the creation and configuration of Virtual LANs (VLANs) on a Cisco Layer 2 switch using Cisco Packet Tracer. Two VLANs — **Student_vlan (VLAN 2)** and **Faculty_vlan (VLAN 3)** — are created, named, and assigned to specific switch interfaces so that end devices connected to those ports become logically isolated into separate broadcast domains despite sharing the same physical switch. The lab is carried out in two stages: first on a single switch, and then the topology is extended with a second switch to verify that VLAN membership and connectivity are preserved across an inter-switch link. Connectivity within each VLAN is verified at every stage using the `ping` command from the command prompt of the connected laptops.

Key learning outcomes covered:

- Creating and naming VLANs using the VLAN database mode on the switch CLI.
- Assigning a range of switch interfaces to a specific VLAN using `switchport access vlan`.
- Verifying VLAN configuration using `show vlan brief`.
- Testing intra-VLAN connectivity between hosts using `ping`.
- Extending the topology with a second switch and replicating the VLAN scheme so that devices in the same VLAN, on different switches, can still communicate.
- Recognising and interpreting common CLI errors (invalid commands) and switch log messages such as `%CDP-4-NATIVE_VLAN_MISMATCH`.

IP Addressing Plan

VLAN	Name	Designed Subnet	Subnet Actually Used in Packet Tr
2	Student_vlan	192.168.1.0 /24	192.138.1.0 /24
3	Faculty_vlan	192.168.2.0 /24	192.138.2.0 /24

Note: The addressing scheme was originally planned as 192.168.1.0/24 and 192.168.2.0/24. During host configuration in Packet Tracer, the third octet was typed as 138 instead of 168 (192.138.x.x), and this addressing was used consistently for all hosts, so all pings shown in this report are performed against the 192.138.x.x range. This is highlighted in Figure 6, where an initial ping to the originally planned address 192.168.1.11 fails, and the ping succeeds only once the actually-configured address 192.138.1.11 is used.

Part 1 — Single-Switch VLAN Configuration

1.1 Initial Topology

A single 2960-24TT switch (**Switch0**) is connected to six laptops. Laptops 0 and 1 are connected to Fa0/1 and Fa0/2 respectively (to become Student_vlan), Laptops 2 and 4 are connected to Fa0/3 and Fa0/4 respectively (to become Faculty_vlan), and Laptop5 is attached with a console cable and used purely to access the switch CLI for configuration.

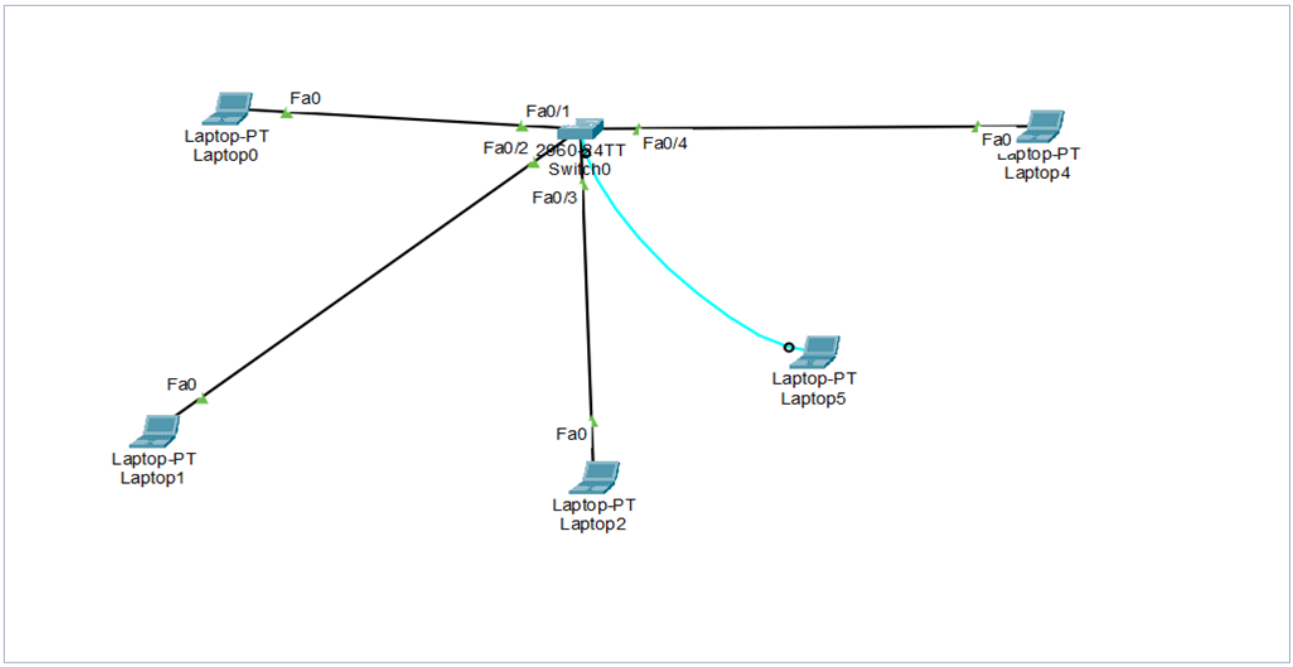


Figure 1 — Initial single-switch topology (Switch0 with Laptop0–Laptop5)

1.2 Creating and Naming the VLANs

The switch is accessed through Laptop5's console connection. From privileged EXEC mode, global configuration mode is entered, and VLAN 2 and VLAN 3 are created and named. Note the first attempt to name VLAN 2 is mistyped as `nmae` instead of `name`, which the IOS correctly rejects as an invalid command before it is re-entered correctly.

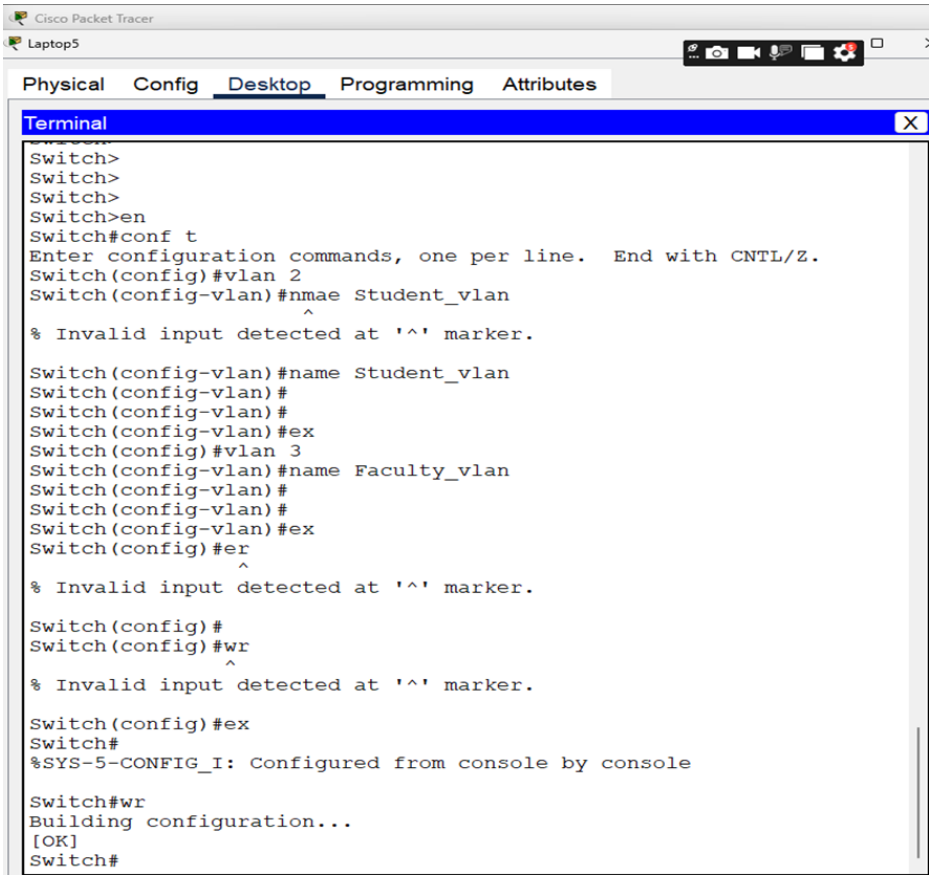


Figure 2 — Console session on Switch0 (via Laptop5) creating VLAN 2 (Student_vlan) and VLAN 3 (Faculty_vlan)

Command Panel — Figure 2 (Switch0 console, transcribed)

```
Switch>
Switch>
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#nmae Student_vlan
^
% Invalid input detected at '^' marker.

Switch(config-vlan)#name Student_vlan
Switch(config-vlan)#
Switch(config-vlan)#
Switch(config-vlan)#ex
Switch(config)#vlan 3
Switch(config-vlan)#name Faculty_vlan
Switch(config-vlan)#
Switch(config-vlan)#
Switch(config-vlan)#ex
Switch(config)#er
^
% Invalid input detected at '^' marker.

Switch(config)#
Switch(config)#wr
^
% Invalid input detected at '^' marker.

Switch(config)#ex
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#
```

1.3 Assigning Access Ports to VLAN 2 (Student_vlan)

Interfaces Fa0/1 and Fa0/2 (connected to Laptop0 and Laptop1) are grouped with `int range` and set to access mode in VLAN 2.

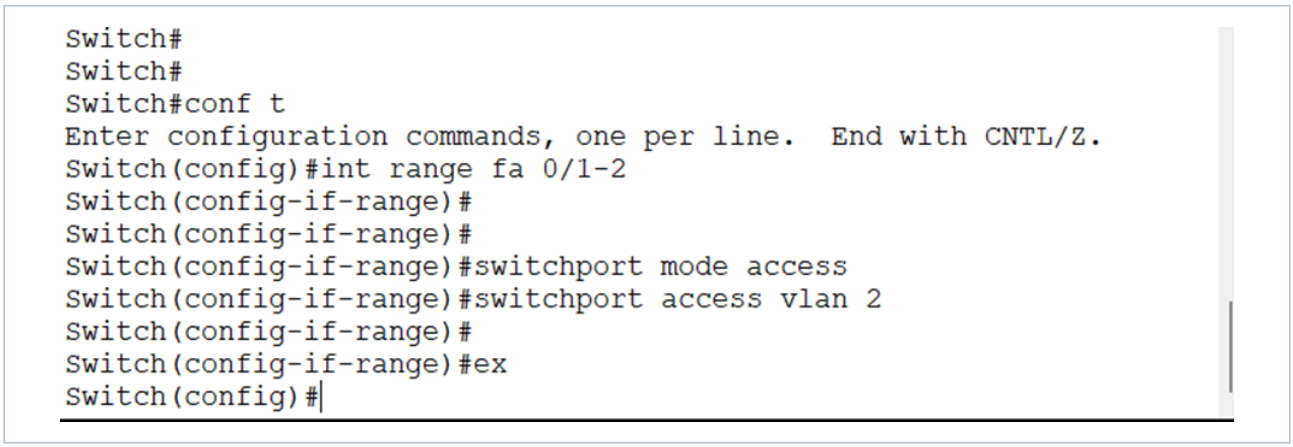


Figure 3 — Fa0/1–Fa0/2 configured as access ports in VLAN 2 (Student_vlan)

Command Panel — Figure 3

```
Switch#
Switch#
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range fa 0/1-2
Switch(config-if-range)#
Switch(config-if-range)#
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 2
Switch(config-if-range)#
Switch(config-if-range)#ex
Switch(config)#
```

1.4 Assigning Access Ports to VLAN 3 (Faculty_vlan)

Interfaces Fa0/3 and Fa0/4 (connected to Laptop2 and Laptop4) are similarly grouped and assigned to VLAN 3.

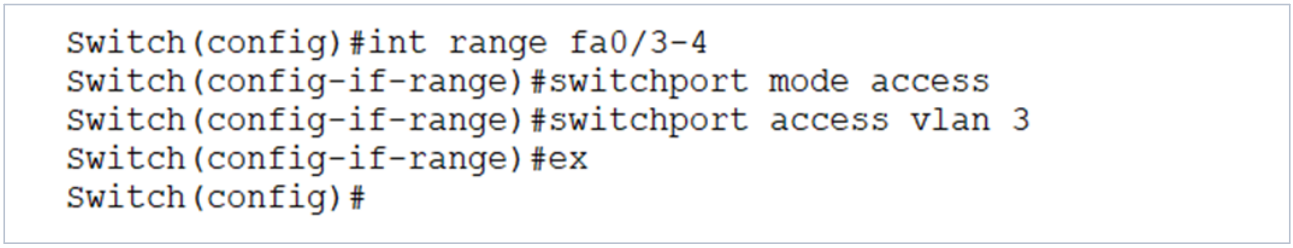


Figure 4 — Fa0/3–Fa0/4 configured as access ports in VLAN 3 (Faculty_vlan)

Command Panel — Figure 4

```
Switch(config)#int range fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 3
Switch(config-if-range)#ex
Switch(config)#
```

1.5 Verifying VLAN Configuration

The `show vlan brief` command confirms that VLAN 2 (Student_vlan) owns ports Fa0/1 and Fa0/2, and VLAN 3 (Faculty_vlan) owns ports Fa0/3 and Fa0/4, while all remaining ports stay in the default VLAN 1.

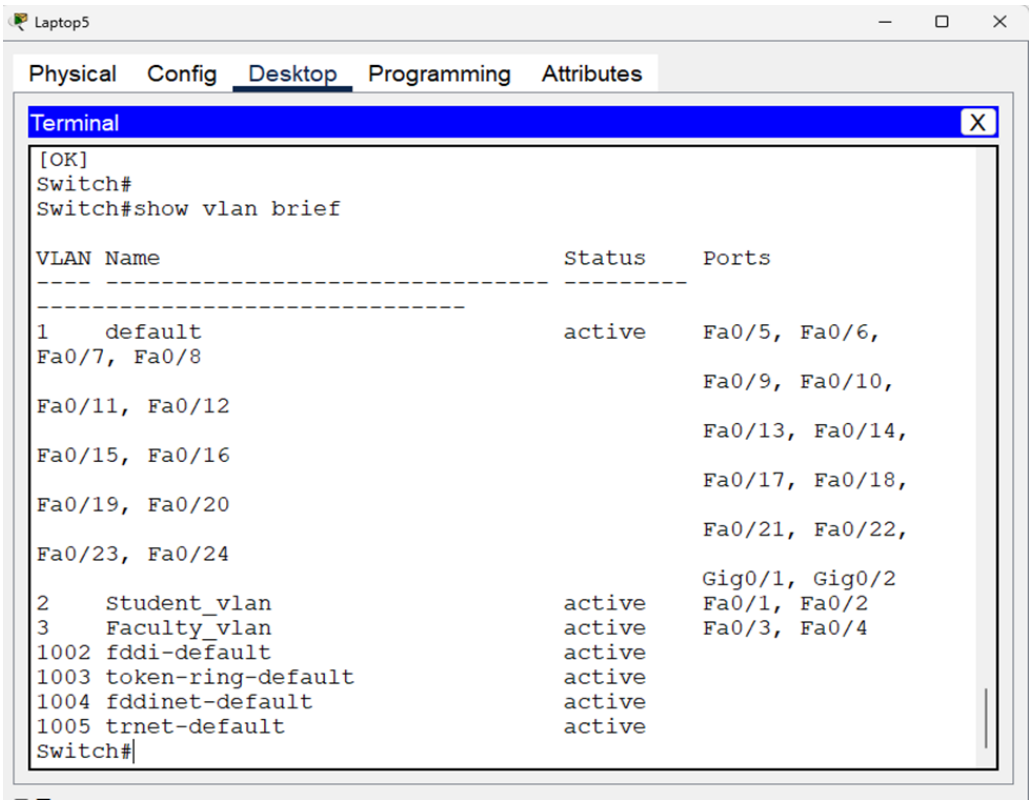


Figure 5 — show vlan brief output confirming Student_vlan and Faculty_vlan port membership

Command Panel — Figure 5

```
[OK]
Switch#
Switch#show vlan brief

VLAN Name Status Ports
---- -------------------------------- --------- -------------------------------
1 default active Fa0/5, Fa0/6, Fa0/7, Fa0/8,
Fa0/9, Fa0/10, Fa0/11, Fa0/12,
Fa0/13, Fa0/14, Fa0/15, Fa0/16,
Fa0/17, Fa0/18, Fa0/19, Fa0/20,
Fa0/21, Fa0/22, Fa0/23, Fa0/24,
Gig0/1, Gig0/2
2 Student_vlan active Fa0/1, Fa0/2
3 Faculty_vlan active Fa0/3, Fa0/4
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
```

1.6 Connectivity Verification — Student_vlan (Laptop0 ↔ Laptop1)

Laptop0 (192.138.1.10) pings Laptop1 (192.138.1.11). The first attempt uses the originally planned address 192.168.1.11 and times out, since no host on the network actually holds that address; once the correct, actually-configured address 192.138.1.11 is used, the ping succeeds.

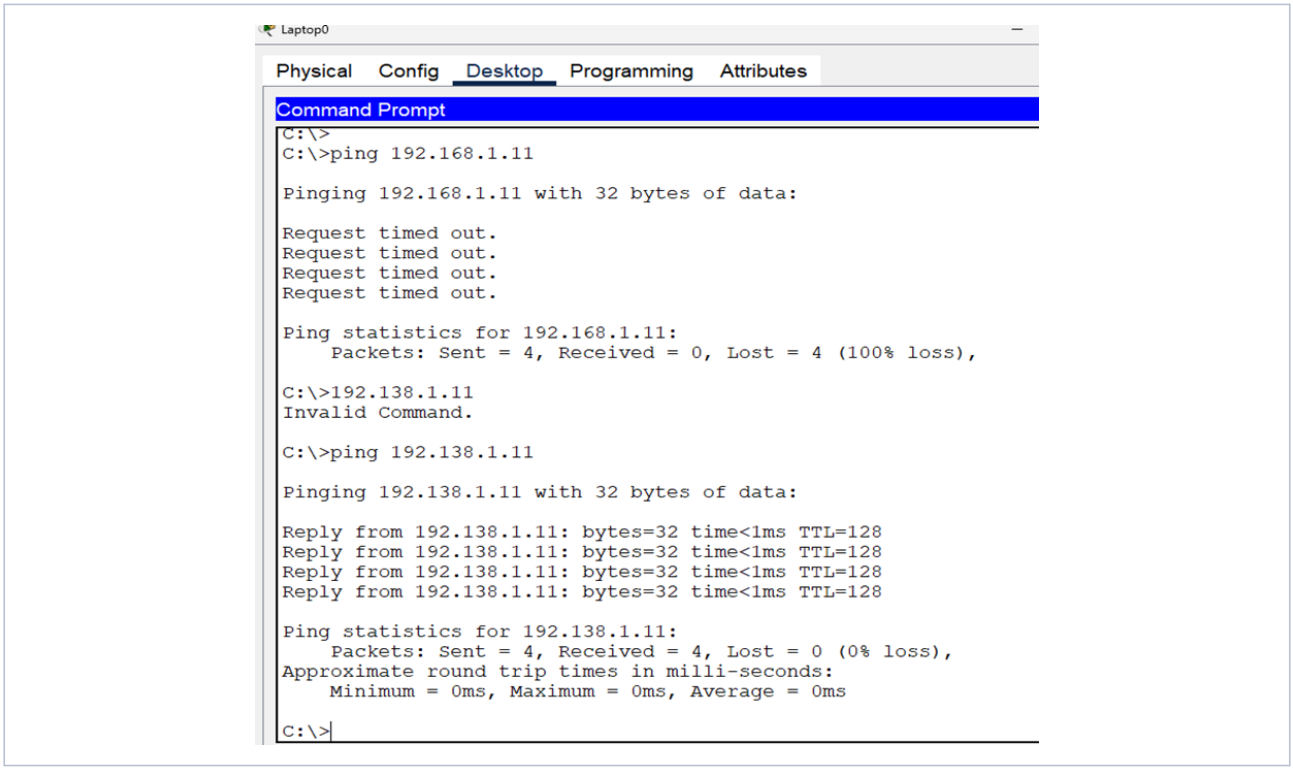


Figure 6 — Laptop0 → Laptop1 ping (fails on the planned 192.168.x address, succeeds on the actual 192.138.x address)

Command Panel — Figure 6 (Laptop0)

```
C:\>
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.11:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>192.138.1.11
Invalid Command.

C:\>ping 192.138.1.11

Pinging 192.138.1.11 with 32 bytes of data:

Reply from 192.138.1.11: bytes=32 time<1ms TTL=128
Reply from 192.138.1.11: bytes=32 time<1ms TTL=128
Reply from 192.138.1.11: bytes=32 time<1ms TTL=128
Reply from 192.138.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.138.1.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

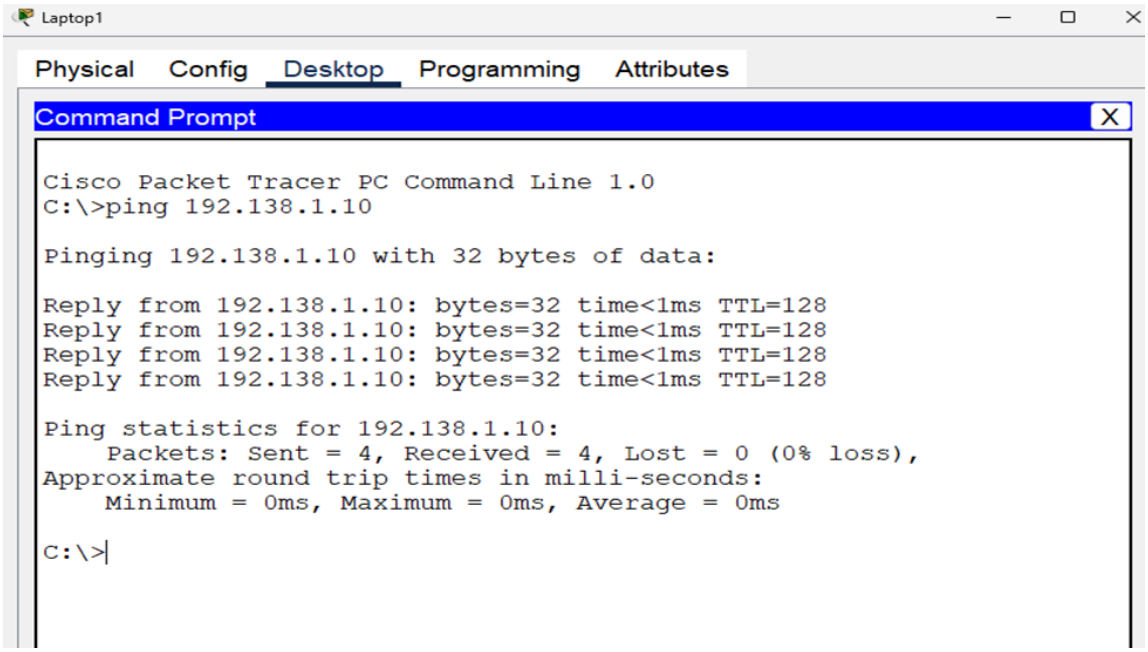


Figure 7 — Reverse test: Laptop1 → Laptop0 (192.138.1.10) — successful

Command Panel — Figure 7 (Laptop1)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.138.1.10

Pinging 192.138.1.10 with 32 bytes of data:

Reply from 192.138.1.10: bytes=32 time<1ms TTL=128
Reply from 192.138.1.10: bytes=32 time<1ms TTL=128
Reply from 192.138.1.10: bytes=32 time<1ms TTL=128
Reply from 192.138.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.138.1.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

1.7 Connectivity Verification — Faculty_vlan (Laptop2 ↔ Laptop4)

Laptop2 (192.138.2.10) and Laptop4 (192.138.2.11) sit on Fa0/3 and Fa0/4, both members of VLAN 3 (Faculty_vlan). Pings in both directions succeed, confirming Faculty_vlan connectivity.

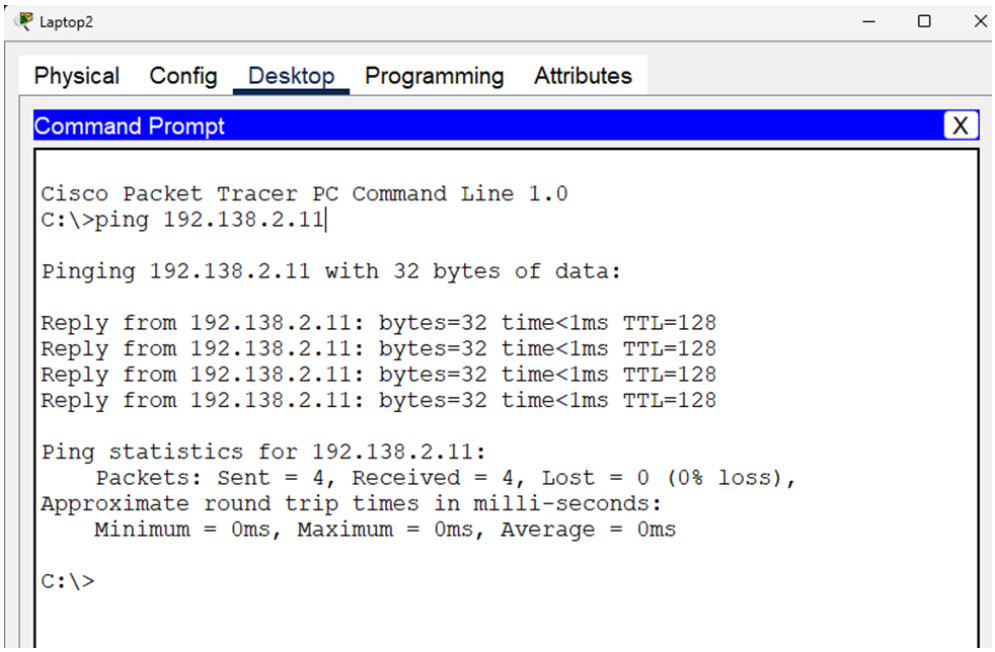


Figure 8 — Laptop2 → Laptop4 (192.138.2.11) — successful

Command Panel — Figure 8 (Laptop2)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.138.2.11

Pinging 192.138.2.11 with 32 bytes of data:

Reply from 192.138.2.11: bytes=32 time<1ms TTL=128
Reply from 192.138.2.11: bytes=32 time<1ms TTL=128
Reply from 192.138.2.11: bytes=32 time<1ms TTL=128
Reply from 192.138.2.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.138.2.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

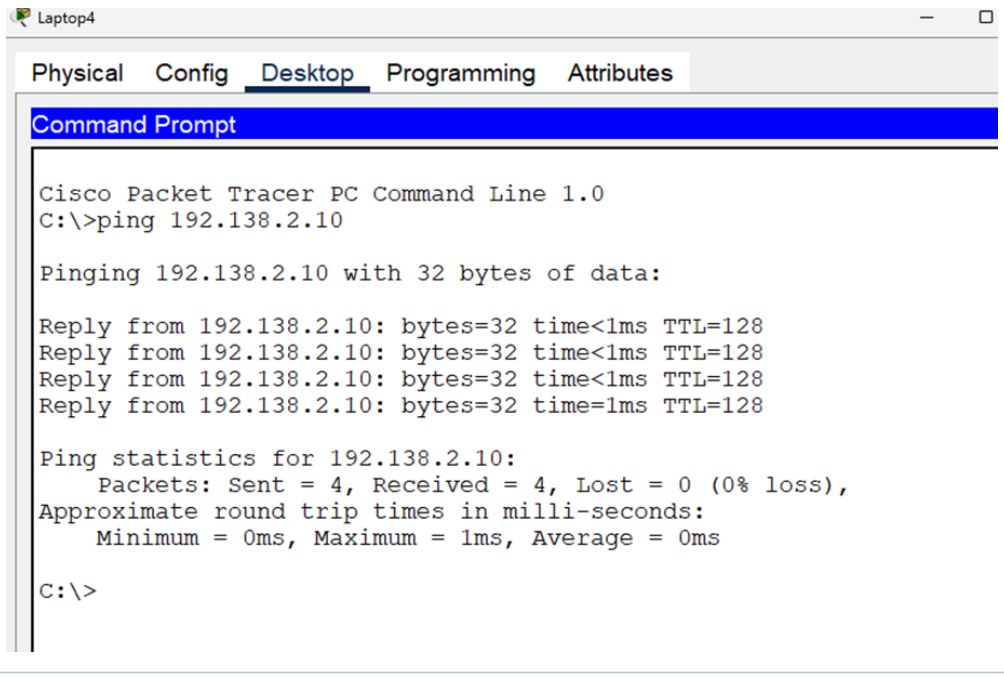


Figure 9 — Reverse test: Laptop4 → Laptop2 (192.138.2.10) — successful

Command Panel — Figure 9 (Laptop4)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.138.2.10

Pinging 192.138.2.10 with 32 bytes of data:

Reply from 192.138.2.10: bytes=32 time<1ms TTL=128
Reply from 192.138.2.10: bytes=32 time<1ms TTL=128
Reply from 192.138.2.10: bytes=32 time<1ms TTL=128
Reply from 192.138.2.10: bytes=32 time=1ms TTL=128

Ping statistics for 192.138.2.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Result: Both VLANs are correctly isolated at Layer 2 while devices within the same VLAN can reach each other freely, confirming that VLAN segmentation on a single switch is working as intended.

Part 2 — Extending the Topology with a Second Switch

To confirm that VLAN membership can span more than one switch, a second switch (**Switch1**) is added with three more laptops: Laptop6 and Laptop7 as data hosts, and Laptop8 attached by console cable for CLI access to Switch1. Switch0 and Switch1 are then linked together so that hosts in the same VLAN on either switch can communicate.

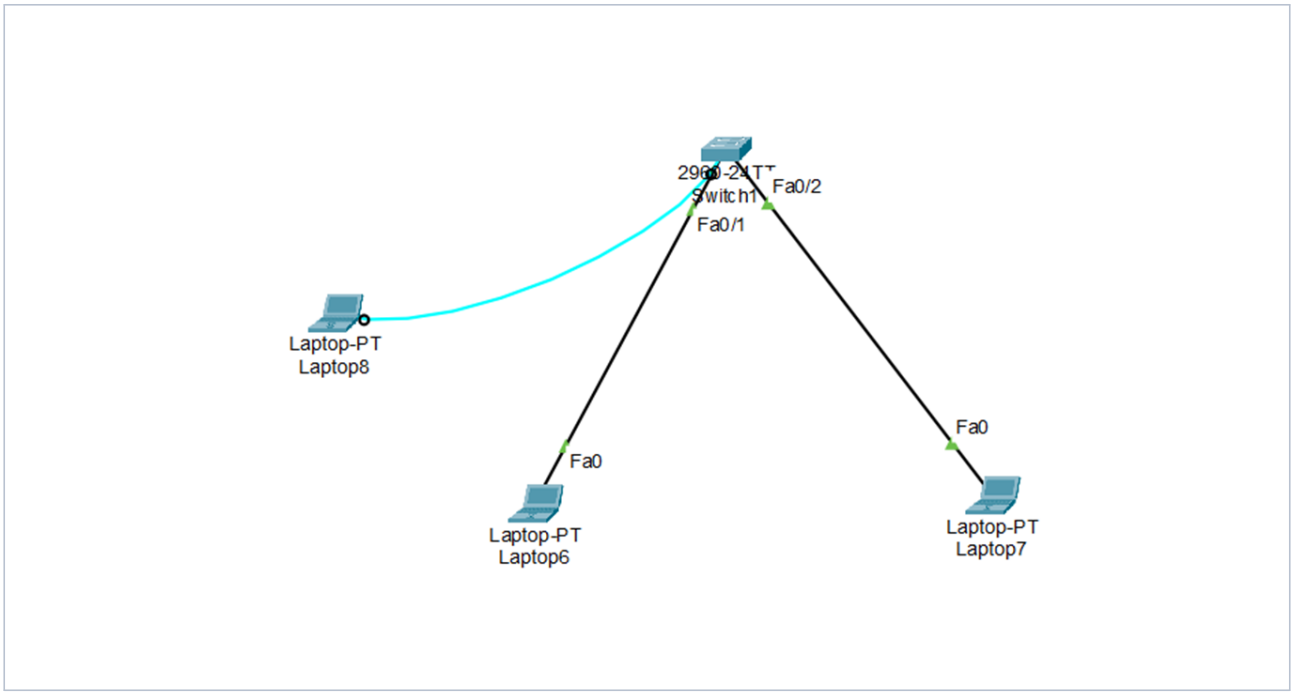


Figure 10 — New segment: Switch1 with Laptop6, Laptop7 (data) and Laptop8 (console access)

2.1 Creating the Same VLANs on Switch1

For the two switches to carry the Student_vlan and Faculty_vlan consistently, VLAN 2 and VLAN 3 are created and named identically on Switch1, mirroring the configuration done on Switch0 in Section 1.2.

```
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 2
Switch(config-vlan)#name Student_vlan
Switch(config-vlan)#ex
Switch(config)#vlan 3
Switch(config-vlan)#name Faculty_vlan
Switch(config-vlan)#ex
Switch(config)#ex
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 11 — VLAN 2 (Student_vlan) and VLAN 3 (Faculty_vlan) created on Switch1

Command Panel — Figure 11 (Switch1 console, via Laptop8)

```
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 2
Switch(config-vlan)#name Student_vlan
Switch(config-vlan)#ex
Switch(config)#vlan 3
Switch(config-vlan)#name Faculty_vlan
Switch(config-vlan)#ex
Switch(config)#ex
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

2.2 Linking the Switches and Extending Student_vlan

Switch0 and Switch1 are cabled together (Switch0 Fa0/5 ↔ Switch1 Fa0/3, and Switch0 Fa0/6 ↔ Switch1 Fa0/4, shown as the dashed inter-switch links) so that VLAN traffic can cross between the two switches. On Switch1, port Fa0/3 (towards Laptop8's data path once reused / towards the Student_vlan side) is placed into VLAN 2, extending Student_vlan onto the second switch.

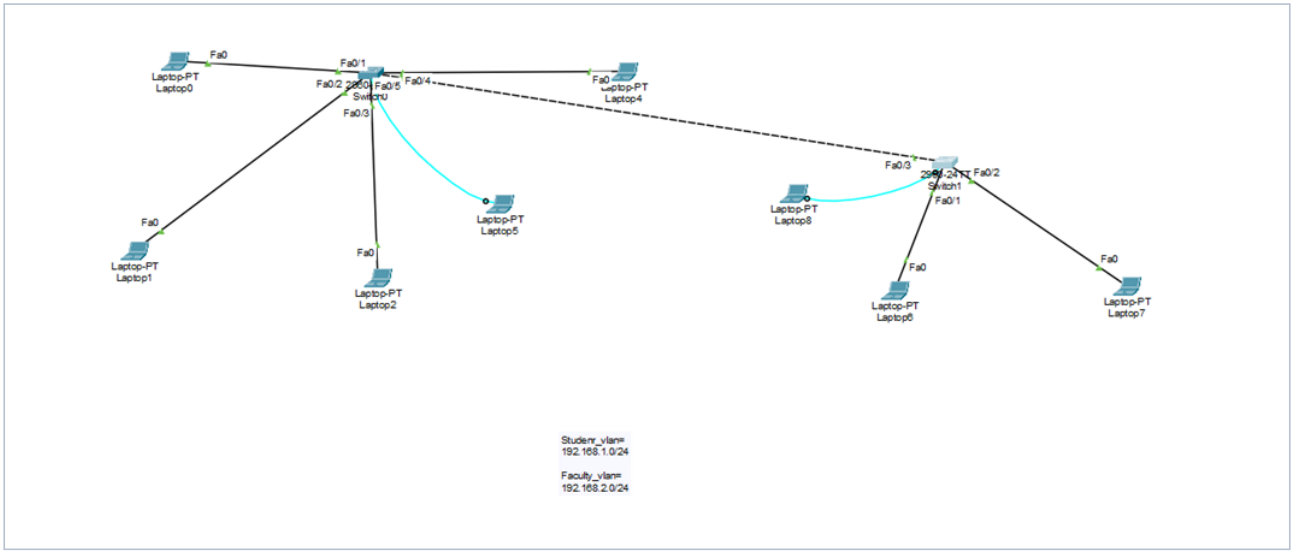


Figure 12 — Combined topology after linking Switch0 and Switch1, with the Student_vlan (192.168.1.0/24) and Faculty_vlan (192.168.2.0/24) subnets labelled

On Switch1, interface Fa0/3 is set to access mode and placed in VLAN 2. Notice the invalid attempt `switchport mode vlan 2` (mode is used to set trunk/access type, not VLAN membership) before the correct command `switchport access vlan 2` is used. The switch log also reports a `%CDP-4-NATIVE_VLAN_MISMATCH` warning, indicating that the native VLAN configured on the Switch0 side of the inter-switch link (VLAN 1) does not match the VLAN configured on this Switch1 port (VLAN 2).

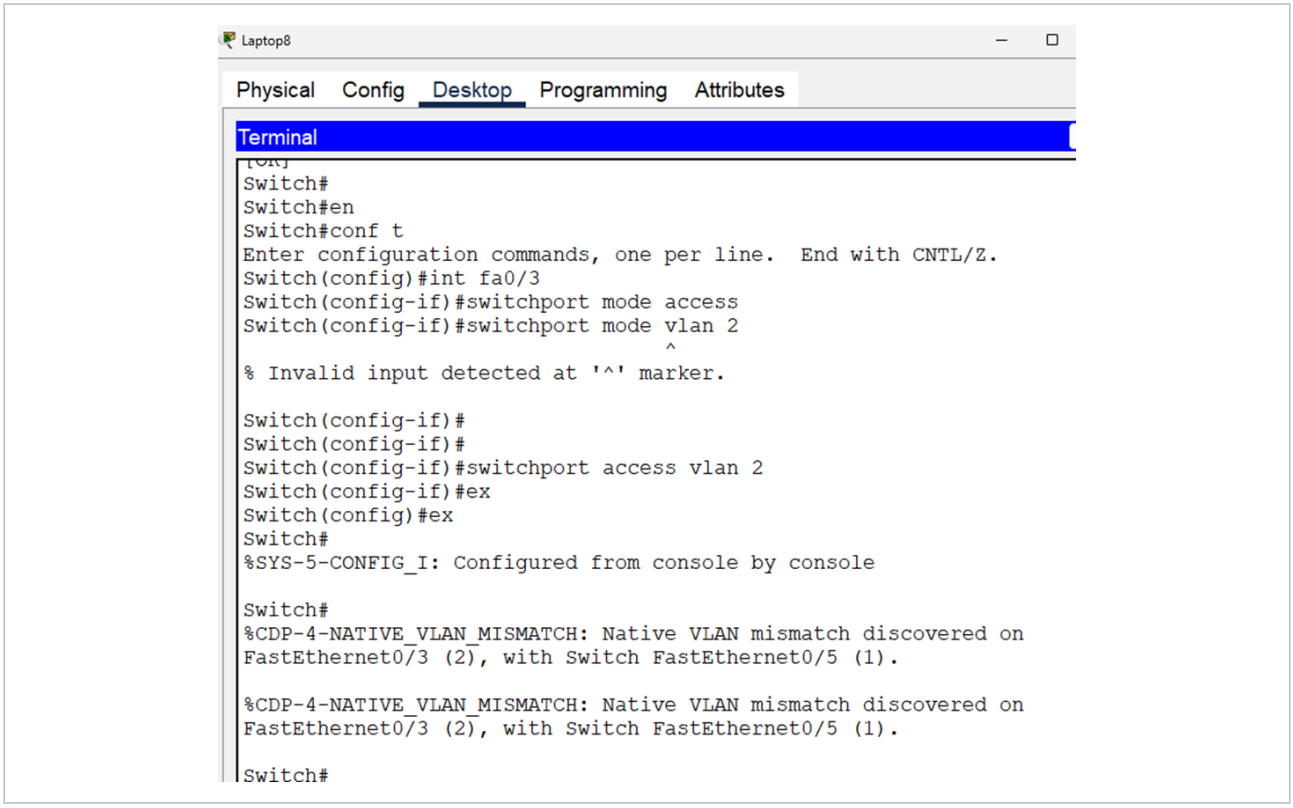


Figure 13 — Switch1 — Fa0/3 assigned to VLAN 2 (Student_vlan); native VLAN mismatch reported on the inter-switch link

Command Panel — Figure 13 (Switch1 console)

```
[OK]
Switch#
Switch#en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport mode vlan 2
^
% Invalid input detected at '^' marker.

Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 2
Switch(config-if)#ex
Switch(config)#ex
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
FastEthernet0/3 (2), with Switch FastEthernet0/5 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
FastEthernet0/3 (2), with Switch FastEthernet0/5 (1).

Switch#
```

2.3 Extending Faculty_vlan Across the Link

The same is repeated for Faculty_vlan: the topology view below highlights the corresponding link and port used for the faculty side, and on Switch0 interface Fa0/1 is set to access mode and placed into VLAN 3, extending Faculty_vlan to the corresponding port. A second native VLAN mismatch message is logged on the other inter-switch link (Fa0/4 ↔ Fa0/6), for the same reason as before.

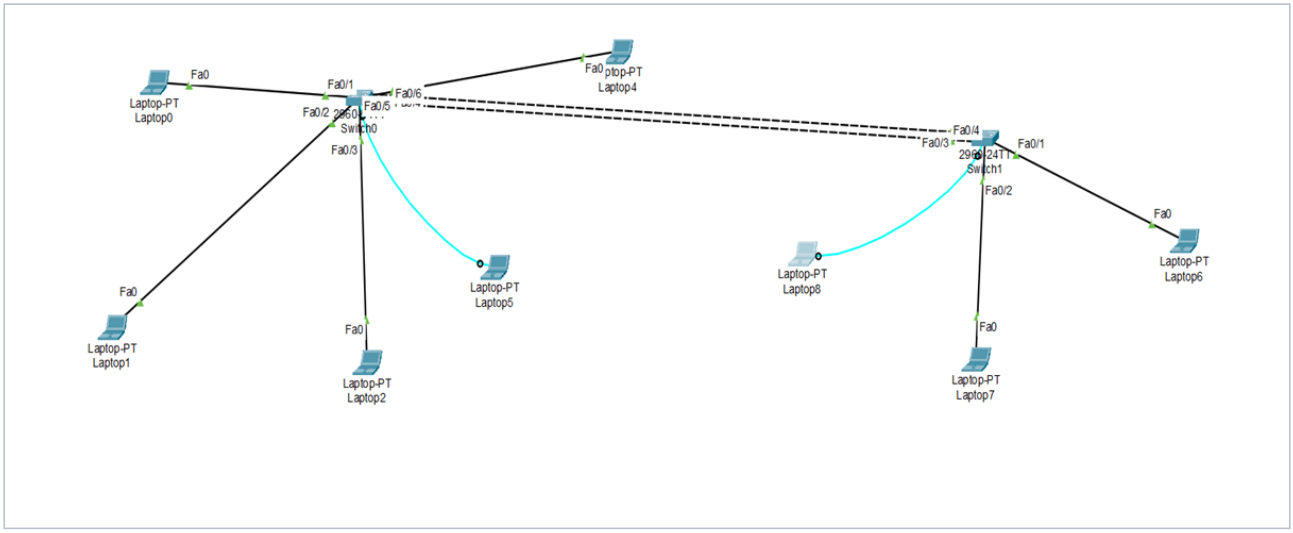


Figure 14 — Full extended topology showing both inter-switch trunk links (Fa0/5–Fa0/3 and Fa0/6–Fa0/4) between Switch0 and Switch1

```
Switch#
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
FastEthernet0/4 (3), with Switch FastEthernet0/6 (1).

% Incomplete command.
Switch(config-if)#
Switch(config-if)#switchport access vlan 3
Switch(config-if)#ex
Switch(config)#ex
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
```

Figure 15 — Switch0 — Fa0/1 assigned to VLAN 3 (Faculty_vlan); native VLAN mismatch reported and configuration saved

Command Panel — Figure 15 (Switch0 console)

```
Switch#
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
FastEthernet0/4 (3), with Switch FastEthernet0/6 (1).

% Incomplete command.
Switch(config-if)#
Switch(config-if)#switchport access vlan 3
Switch(config-if)#ex
Switch(config)#ex
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
```

2.4 Connectivity Verification on the Extended Topology

Finally, connectivity is re-tested from the Student_vlan and Faculty_vlan sides after the topology extension. Laptop1 first pings a host address (192.138.1.12) that is not yet reachable and times out, then succeeds once the correct host/port assignment is in place. Laptop4 successfully pings both 192.138.2.10 and the newly reachable 192.138.2.12, confirming Faculty_vlan connectivity across the extended, two-switch topology.

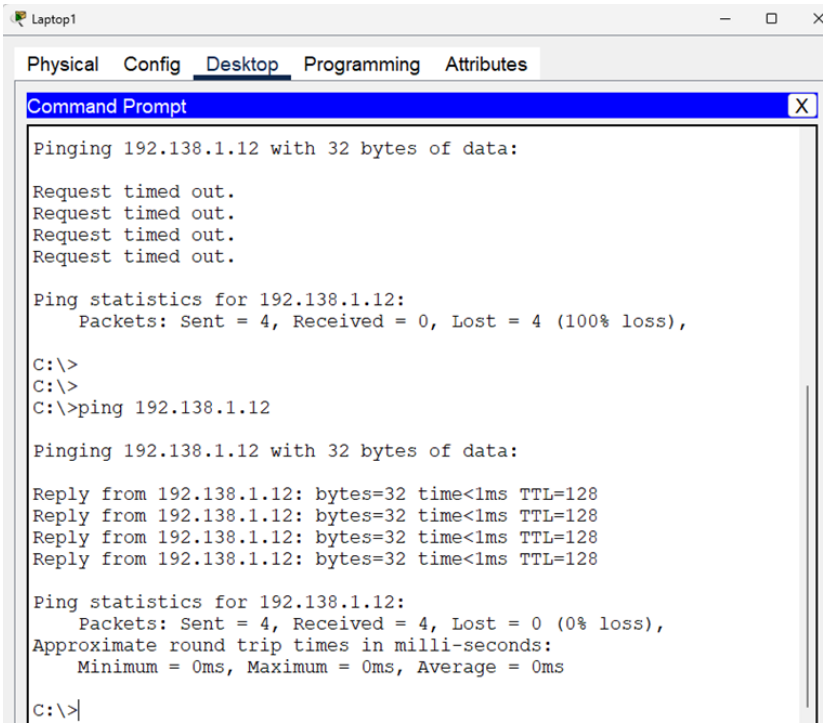


Figure 16 — Laptop1 ping test on the extended topology (192.138.1.12) — times out, then succeeds

Command Panel — Figure 16 (Laptop1)

```
Pinging 192.138.1.12 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.138.1.12:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
C:\>
C:\>ping 192.138.1.12

Pinging 192.138.1.12 with 32 bytes of data:

Reply from 192.138.1.12: bytes=32 time<1ms TTL=128
Reply from 192.138.1.12: bytes=32 time<1ms TTL=128
Reply from 192.138.1.12: bytes=32 time<1ms TTL=128
Reply from 192.138.1.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.138.1.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

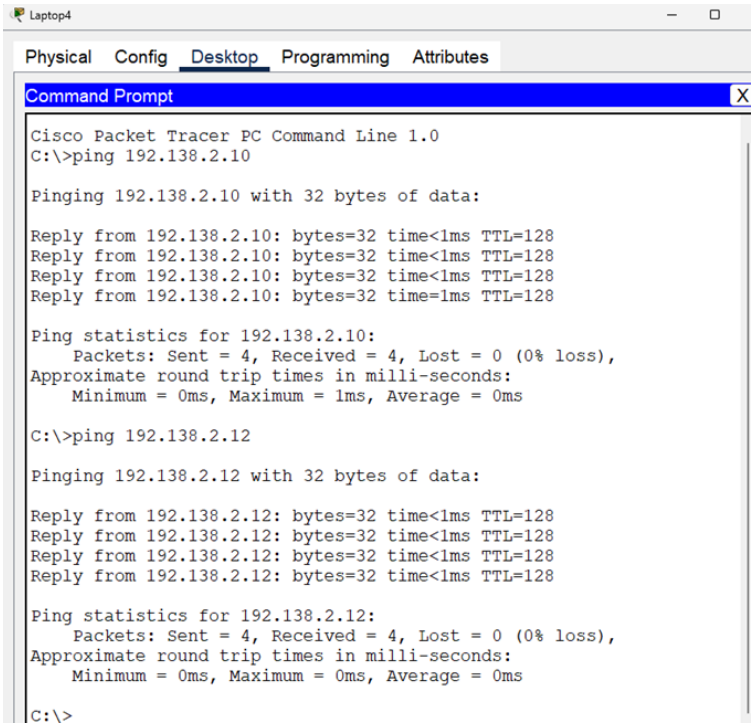


Figure 17 — Laptop4 ping test on the extended topology (192.138.2.10 and 192.138.2.12) — both successful

Command Panel — Figure 17 (Laptop4)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.138.2.10

Pinging 192.138.2.10 with 32 bytes of data:

Reply from 192.138.2.10: bytes=32 time<1ms TTL=128
Reply from 192.138.2.10: bytes=32 time<1ms TTL=128
Reply from 192.138.2.10: bytes=32 time<1ms TTL=128
Reply from 192.138.2.10: bytes=32 time=1ms TTL=128

Ping statistics for 192.138.2.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.138.2.12

Pinging 192.138.2.12 with 32 bytes of data:

Reply from 192.138.2.12: bytes=32 time<1ms TTL=128
Reply from 192.138.2.12: bytes=32 time<1ms TTL=128
Reply from 192.138.2.12: bytes=32 time<1ms TTL=128
Reply from 192.138.2.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.138.2.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Observations and Troubleshooting Notes

- **Command typos are self-correcting in IOS:** mistyped commands such as `nmae` instead of `name`, or `switchport mode vlan 2` instead of `switchport access vlan 2`, are rejected by the CLI with an 'Invalid input detected' message and a caret pointing at the offending character; the correct command was simply re-entered immediately after.
- **Addressing discrepancy:** hosts were actually configured on the 192.138.x.x range rather than the originally planned 192.168.x.x range. This was caught directly by a failed ping (Figure 6) and does not affect the validity of the VLAN configuration itself, since VLANs operate at Layer 2 independently of the IP addressing scheme.
- **Native VLAN mismatch:** once the inter-switch links were brought up with access ports assigned to VLAN 2/3 on either end rather than being configured as trunk links with a matching native VLAN, Packet Tracer logged `%CDP-4-NATIVE_VLAN_MISMATCH` messages. This is expected because each inter-switch port was left as an access port in a different VLAN from its peer, rather than the default VLAN 1 on both ends; despite the warning, traffic for the matching VLAN still passed and connectivity tests (Figures 16 and 17) succeeded.
- **Verification strategy:** at every stage, connectivity was verified in both directions (e.g. Laptop0→Laptop1 and Laptop1→Laptop0) rather than a single direction, which is good practice since it also confirms that ARP resolution and the return path work correctly.

Conclusion

This lab successfully demonstrated end-to-end VLAN configuration in Cisco Packet Tracer: creating and naming VLANs, assigning access ports, verifying the configuration through `show vlan brief`, and confirming Layer 2 connectivity through `ping`. The topology was then extended with a second switch, and the same VLAN scheme was replicated so that Student_vlan and Faculty_vlan traffic could correctly traverse both switches. All connectivity tests within each VLAN passed, confirming that VLAN segmentation was correctly implemented and preserved across the multi-switch topology, despite a few CLI typos and native VLAN mismatch warnings encountered and resolved along the way.